Week 6: Conditional Generation

Theme:	Don't just generate random digits. Generate the digit you want.
Time Commitment:	9–11 hours
Suggested Split:	3h reading, 7h coding/training/experimenting, 1h writeup

Learning Objectives

By the end of this week, you should be able to:

- Implement class-conditional generation using label embeddings
- Implement and tune classifier-free guidance (CFG)
- Articulate the quality vs. diversity trade-off controlled by guidance scale
- Lay groundwork for text conditioning (foundation for Stable Diffusion)

Primary Resources (Must-Do)

[PAPER] Classifier-Free Guidance Paper (Ho & Salimans, 2022)

Type: Research Paper | **Time:** 60 min | **Difficulty:** Intermediate-Advanced
URL: https://arxiv.org/pdf/2207.12598

Why this resource: Short, accessible paper introducing the technique used in essentially every modern conditional diffusion model (Stable Diffusion, DALL-E, Imagen). The math is straightforward.

Focus on: The training algorithm with random conditioning dropout, and the sampling formula with the guidance scale.

[BLOG/DOCS] Sander Dieleman: "Guidance: a cheat code for diffusion models"

Type: Blog Post | **Time:** 45 min | **Difficulty:** Intermediate
URL: https://sander.ai/2022/05/26/guidance.html

Why this resource: The most intuitive explanation of CFG online. Sander Dieleman is a Google DeepMind researcher who explains diffusion better than almost anyone.

Focus on: The geometric interpretation of guidance — it's pushing predictions in the direction of higher conditional likelihood.

[BLOG/DOCS] Hugging Face Diffusion Course Unit 2: Fine-Tuning and Guidance

Type: Tutorial + Code | **Time:** 60–90 min | **Difficulty:** Intermediate
URL: https://huggingface.co/learn/diffusion-course/unit2/1

Why this resource: Practical, hands-on coverage of conditioning with code you can run. Use this as a coding companion alongside your own implementation.

[PAPER] Stable Diffusion Paper (Rombach et al., 2022) — Section 3 only

Type: Research Paper (selected sections) | **Time:** 30 min | **Difficulty:** Advanced
URL: https://arxiv.org/pdf/2112.10752

Why this resource: Just to see how text conditioning is plumbed in via cross-attention. You won't implement this fully, but understanding it sets you up for any future text-to-image work.

Focus on: Section 3.3 (Conditioning Mechanisms). Skip the latent space discussion for now.

This Week's Deliverable

Extend your diffusion model to support conditional generation. Code must include:

- A label embedding layer that converts integer class labels to dense vectors
- Label embeddings concatenated or added to timestep embeddings (you decide; document choice)
- Random label dropout during training (10–20% replaced with a "null" token) to enable CFG
- CFG-aware sampling with adjustable guidance scale w
- **Sample grid 1:** Same noise seed, different class labels (shows control)
- **Sample grid 2:** Same class label, different guidance scales: $w = 1, 3, 5, 10$ (shows the quality-diversity tradeoff)
- Brief writeup analyzing the visual differences across guidance scales

Key Concepts Checklist

You should be able to confidently answer all of the following:

- Why must we randomly drop the conditioning during training?
- Write out the CFG sampling formula. What does it geometrically mean?
- Why does increasing w improve sample quality but reduce diversity?
- What's the difference between classifier guidance (older) and classifier-free guidance?
- How would you extend this to text conditioning? What changes?
- What happens if $w = 0$? What happens if $w = 1$? What happens if $w = 100$?

Stretch Resources (Optional)

- **Replace class labels with CLIP text embeddings** for true text-to-image (small scale, like "sad face" / "happy face" on emoji data)
- **Read the GLIDE paper** (Nichol et al., 2022): https://arxiv.org/abs/2112.10741 — combines CLIP guidance with diffusion
- **Implement classifier guidance** (the older Dhariwal & Nichol approach) for comparison
- Visualize the noise prediction direction with and without guidance — they literally point in different directions

Debugging & Reference

- **If CFG has no effect:** You probably forgot to randomly drop conditioning during training
- **If high w produces solid color images:** Guidance is over-saturating; cap w around 7–10 for typical models

- **Sign error check:** CFG formula is $\hat{\epsilon} = \epsilon_{\text{uncond}} + w \cdot (\epsilon_{\text{cond}} - \epsilon_{\text{uncond}})$
- **Null token:** Many implementations use class label -1 or a learned null embedding; both work